

A Control-Anchored Mean-Field Neural Differential Equation Formulation for P4-to-P60 Perturb-seq Dynamics

Abstract

We formulate the $P4 \rightarrow P60$ perturb-seq problem as inference of a perturbation-indexed family of finite-measure-valued stochastic processes on a latent cell-state space. The biological motivation is that a perturbation can alter at least three coupled aspects of tumour evolution: directional progression through cell-state space, stochastic heterogeneity around that progression, and net expansion or depletion of the perturbed population. Because single-cell RNA sequencing is destructive, the observed data are snapshot distributions rather than tracked single-cell trajectories. This motivates a weighted stochastic differential equation formulation in a shared latent space. For each perturbation $g \in \mathcal{G}$, the observed $P4$ cells and guide-abundance scale define an initial finite measure $\mu_0^g = M_g^0 p_0^g$, while the observed $P60$ cells and guide-abundance scale define a terminal finite measure $\nu_1^g = M_g^1 p_1^g$. The model is control anchored: control perturbations are represented by the zero embedding and therefore follow the baseline continuation law of the same coefficient networks. The model is also mean-field coupled: all perturbation-specific populations jointly determine a low-dimensional tissue context, and this context feeds back into the perturbation-conditioned dynamics. Mathematically, the resulting system is a reaction–drift–diffusion McKean–Vlasov model in latent space, implemented as a weighted neural stochastic differential equation with three separate control-anchored neural networks for directional movement, stochastic dispersion, and multiplicative growth or depletion. The predicted terminal finite measures are fitted to data with a single entropic unbalanced optimal-transport discrepancy, because terminal geometry and terminal abundance are two aspects of the same endpoint measure rather than two independent targets. Coarse cell-population labels may be used to build interpretable tissue-context summaries, but they are not required for the core endpoint supervision. The learned object is an effective interval generator for the single observed interval $P4 \rightarrow P60$; it is not claimed to be the unique microscopic biological law between the two snapshots.

Contents

1	Biological background and modeling target	3
1.1	Biological setting	3
1.2	Why the data imply a population model rather than a trajectory-matching model . .	3
1.3	Why control anchoring is natural	4
1.4	Why mean-field tissue context is biologically plausible	4
1.5	What the model should estimate	4
2	Explicit introduction to the SDE	4
2.1	What an SDE is	4
2.2	Why an ODE alone is not enough	5

2.3	Why we also need a growth equation	5
2.4	The weighted SDE we actually want	5
2.5	From a single weighted particle to a population measure	6
3	Notation table	6
4	Observed data and endpoint object	7
4.1	Normalized time	7
4.2	Latent state space	8
4.3	Perturbation-indexed single-cell observations	8
4.4	Guide-abundance masses and finite endpoint measures	8
4.5	Optional interpretable tissue summaries	9
5	Assumptions	9
6	Control anchoring and mean-field tissue context	9
6.1	Control perturbations and perturbation embeddings	9
6.2	Joint population and total mass	10
6.3	Generic mean-field context	10
6.4	One interpretable choice of context map	10
6.5	Time embedding and common network input	11
7	Explicit control-anchored three-network weighted SDE	11
7.1	Why three separate networks	11
7.2	Drift network: directional movement	11
7.3	Dispersion network: stochastic spread	12
7.4	Growth network: expansion or depletion	12
7.5	Exact control anchor	12
7.6	Particle system	13
7.7	Population measure induced by the particle system	13
7.8	Weak measure form	13
7.9	Regularity conditions	13
8	Explicit details behind the loss function	14
8.1	Why the loss must act on finite measures	14
8.2	Derived endpoint summaries	14
8.3	Generalized KL divergence for finite measures	14
8.4	Entropic unbalanced optimal transport	15
8.5	Debiased unbalanced Sinkhorn divergence	15
8.6	Primary endpoint loss	15
8.7	Optional auxiliary composition diagnostic	15
9	Regularization and identifiability	16
9.1	Embedding shrinkage	16
9.2	Perturbation-modulation regularization	16
9.3	Dispersion regularization	16
9.4	Network parameter regularization	16
9.5	Why strong regularization is necessary	16

10 Full training objective	17
11 Numerical approximation and training	17
11.1 Euler–Maruyama approximation	17
11.2 Context update	17
11.3 Discrete terminal measures	17
11.4 Practical training loop	18
12 Biological outputs and counterfactuals	18
12.1 Predicted terminal finite measure	18
12.2 Predicted abundance effect	18
12.3 Predicted terminal state distribution	19
12.4 Predicted terminal coarse composition	19
12.5 Intrinsic versus context-mediated effect	19
12.6 Control-referenced perturbation effect	19
13 What this model is, and is not	19
13.1 It is a mean-field interaction model	19
13.2 It is not a full mean-field game	19
13.3 It estimates an effective interval generator	19
13.4 It supports intervention-conditional interpretation, not unrestricted causal claims . .	20
14 Concise summary	20

1 Biological background and modeling target

1.1 Biological setting

The biological problem is to understand how a perturbation reshapes tumour evolution over the observed interval

$$P4 \longrightarrow P60.$$

At a biological level, a perturbation can affect at least three coupled mechanisms.

First, it can change the *direction* in which cells move through transcriptomic state space. A perturbation may promote progression toward one epithelial program, block another program, or redirect cells into a different branch of the state manifold.

Second, it can change the amount of *heterogeneity* or unresolved stochastic variation around that dominant direction. Even if two perturbations have similar average trends, one may produce a broader spread of cell states and another may confine the population more tightly.

Third, it can change the *net abundance* of the perturbed population. A perturbation may confer a selective advantage, cause depletion, alter survival, or change proliferative capacity.

These three effects—movement, stochastic spread, and net growth or loss—are precisely the three components that motivate the mathematical structure of the model below.

1.2 Why the data imply a population model rather than a trajectory-matching model

Single-cell RNA sequencing is destructive. One does not observe the same physical cell at $P4$ and again at $P60$. Therefore, the data do not provide directly tracked cell lineages in state space. Instead, for each perturbation, one observes:

- a collection of cell states at $P4$,
- a collection of cell states at $P60$,
- and guide-abundance information that provides a relative mass scale.

This means that the natural object of inference is not a set of matched cell-to-cell trajectories. The natural object is a *population dynamics model* that evolves a perturbation-specific distribution or finite measure from the $P4$ snapshot to the $P60$ snapshot.

1.3 Why control anchoring is natural

The experimental design contains control perturbations, such as non-targeting guides. These controls provide a natural reference continuation law. Rather than learning a completely unrelated high-capacity dynamics for each perturbation, it is more statistically efficient and biologically interpretable to model each perturbation as a deviation from a shared control law. This is the reason for introducing a perturbation embedding a_g with the exact anchor

$$a_g = 0 \quad \text{for controls.}$$

1.4 Why mean-field tissue context is biologically plausible

A perturbation can act cell-intrinsically, but the evolving tissue composition can also feed back on cell behavior. In a tumour setting, the current mixture of epithelial states, immune states, and signaling-related features can alter the effective continuation law. This motivates a mean-field interaction layer: the coefficient fields are allowed to depend not only on the current cell state and perturbation identity, but also on low-dimensional summaries of the current global population.

1.5 What the model should estimate

The target is a perturbation-conditioned family of evolving finite measures

$$\left\{ \mu_t^g : t \in [0, 1], g \in \mathcal{G} \right\},$$

where $t = 0$ corresponds to $P4$ and $t = 1$ corresponds to $P60$. The same model should explain both:

- (i) where the perturbed population ends up in latent state space, and
- (ii) how much total perturbation-specific mass survives, expands, or depletes.

2 Explicit introduction to the SDE

2.1 What an SDE is

A stochastic differential equation describes how a random state evolves over infinitesimal time. If $Z_t \in \mathbb{R}^d$ denotes a latent cell state, a generic SDE has the form

$$dZ_t = b(Z_t, t) dt + \sigma(Z_t, t) dW_t, \tag{1}$$

where:

- $b(Z_t, t)$ is the *drift*, the deterministic infinitesimal direction of motion;

- $\sigma(Z_t, t)$ is the *diffusion map*, which scales stochastic fluctuations;
- W_t is a Brownian motion, the canonical continuous-time noise process.

Over a small time interval Δt , [equation \(1\)](#) can be read heuristically as

$$Z_{t+\Delta t} - Z_t \approx b(Z_t, t) \Delta t + \sigma(Z_t, t) \sqrt{\Delta t} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I). \quad (2)$$

Thus, the drift term determines the average direction of motion, while the diffusion term determines the size and orientation of stochastic spread around that direction.

2.2 Why an ODE alone is not enough

A deterministic ODE would only include the drift term,

$$dZ_t = b(Z_t, t) dt.$$

That is too restrictive here. In the present problem, the latent state does not capture all biological variables, and the observed data consist only of sparse destructive snapshots. Therefore, a stochastic dispersion term is not merely a nuisance; it is part of the effective interval-level description. It absorbs unresolved heterogeneity, latent-state incompleteness, and intrinsically variable continuation around the main direction of motion.

2.3 Why we also need a growth equation

The SDE in [equation \(1\)](#) moves particles through latent state space, but it does not by itself account for perturbation-specific expansion or depletion. To model abundance changes, we attach a positive weight w_t to each particle and write

$$\frac{d}{dt} \log w_t = r(Z_t, t), \quad (3)$$

where r is a scalar growth or reaction field.

This has the explicit solution

$$w_t = \exp\left(\int_0^t r(Z_s, s) ds\right). \quad (4)$$

If $r > 0$ along the trajectory, the particle weight grows; if $r < 0$, the weight shrinks.

2.4 The weighted SDE we actually want

For perturbation-conditioned tumour evolution, the natural conceptual model is therefore

$$dZ_t = b_g(Z_t, t; \mu_t) dt + \sigma_g(Z_t, t; \mu_t) dW_t, \quad (5)$$

together with

$$\frac{d}{dt} \log w_t = r_g(Z_t, t; \mu_t). \quad (6)$$

The perturbation index g appears because different perturbations modify the continuation law. The dependence on μ_t appears because the current tissue context can feed back on the dynamics through a mean-field summary of the evolving population.

2.5 From a single weighted particle to a population measure

A single weighted particle does not represent the full perturbation-specific population. To represent the population, we initialize many particles at the observed $P4$ cells and then evolve them jointly. The perturbation-specific population at time t is the weighted empirical measure

$$\mu_t^g = \frac{M_g^0}{n_g^0} \sum_{i=1}^{n_g^0} w_t^{g,i} \delta_{Z_t^{g,i}}. \quad (7)$$

This measure-valued object is the main mathematical prediction of the model. It contains both the terminal state allocation and the terminal perturbation mass.

3 Notation table

Table 1: Core notation used throughout the formulation.

Symbol	Type	Meaning
$\mathcal{G}$	finite set	Set of perturbations.
$\mathcal{G}_{\text{ctrl}}$	subset of $\mathcal{G}$	Set of control perturbations.
g, h	elements of $\mathcal{G}$	Perturbation indices.
$t \in [0, 1]$	scalar	Normalized time, with $t = 0$ for $P4$ and $t = 1$ for $P60$.
$\tau(t)$	scalar	Optional map from normalized time to physical days.
$\mathcal{Z} \subseteq \mathbb{R}^d$	space	Shared latent cell-state space.
x_{gi}, y_{gj}	vectors in $\mathcal{Z}$	Observed latent states of cell i at $P4$ and cell j at $P60$ under perturbation g .
X_g, Y_g	finite sets	Observed latent cell-state collections at $P4$ and $P60$ under perturbation g .
n_g^0, n_g^1	integers	Numbers of observed $P4$ and $P60$ cells under perturbation g .
p_0^g, p_1^g	probability measures	Empirical probability measures of the observed latent states at $P4$ and $P60$.
M_g^0, M_g^1	positive scalars	Relative perturbation masses on the guide-abundance scale at $P4$ and $P60$.
μ_0^g, ν_1^g	finite measures	Initial and observed terminal finite measures, $\mu_0^g = M_g^0 p_0^g$ and $\nu_1^g = M_g^1 p_1^g$.
μ_t^g	finite measure	Predicted perturbation-specific population measure at time t .
$\bar{\mu}_t$	finite measure	Total population measure $\sum_{g \in \mathcal{G}} \mu_t^g$.
$\bar{M}(t)$	positive scalar	Total mass $\bar{\mu}_t(\mathcal{Z})$.
a_g	vector in $\mathbb{R}^r$	Perturbation embedding; controls satisfy $a_g = 0$.
ω	map $\mathcal{Z} \rightarrow \Delta^{K-1}$	Optional coarse membership map used for interpretable context summaries.
ϕ	map $\mathcal{Z} \rightarrow \mathbb{R}^J$	Optional mediator or signaling feature map.

Symbol	Type	Meaning
q_t	vector in Δ^{K-1}	Optional global coarse composition summary.
s_t	vector in $\mathbb{R}^J$	Optional global mediator summary.
c_t	vector in $\mathbb{R}^L$	Low-dimensional mean-field tissue-context vector.
$\gamma(t)$	vector in $\mathbb{R}^{d_t}$	Time embedding.
$u_t(z; \mu_t)$	vector	Shared network input built from latent state, time, and tissue context.
b_g	vector field	Drift field governing directional movement.
σ_g	matrix field	Diffusion map governing stochastic dispersion.
r_g	scalar field	Growth or reaction field governing multiplicative expansion or depletion.
$W_t^{g,i}$	Brownian motion	Stochastic forcing for particle i under perturbation g .
$Z_t^{g,i}$	stochastic process	Latent state trajectory of particle i under perturbation g .
$w_t^{g,i}$	positive process	Weight of particle i under perturbation g .
$\hat{M}_g^1$	scalar	Predicted terminal mass $\mu_1^g(\mathcal{Z})$.
$\hat{p}_1^g$	probability measure	Predicted terminal normalized state distribution $\mu_1^g / \hat{M}_g^1$.
$\hat{q}_{g,1}$	vector in Δ^{K-1}	Optional predicted terminal coarse composition.
$\sigma_{\min}$	positive scalar	Lower bound used in the diffusion parameterization.
$r_{\max}$	positive scalar	Bound used in the growth parameterization.
ε	positive scalar	Entropic regularization parameter in unbalanced OT.
τ	positive scalar	Marginal-relaxation parameter in unbalanced OT.
$\text{UOT}_{\varepsilon, \tau}$	functional	Entropically regularized unbalanced OT cost on finite measures.
$\text{SD}_{\varepsilon, \tau}^{\text{unb}}$	functional	Debiased unbalanced Sinkhorn divergence on finite measures.
$\theta_b, \theta_\sigma, \theta_r, \theta_c$	parameters	Parameters of the drift, dispersion, growth, and context networks.

4 Observed data and endpoint object

4.1 Normalized time

We identify

$$t = 0 \equiv P4, \quad t = 1 \equiv P60.$$

If one wishes to connect normalized time to physical days, one may set

$$\tau(t) = 4 + 56t,$$

so that $\tau(0) = 4$ and $\tau(1) = 60$.

4.2 Latent state space

Let

$$\mathcal{Z} \subseteq \mathbb{R}^d$$

denote a common latent cell-state space obtained from pooled *P4* and *P60* transcriptomes. This latent space may be PCA, scVI, or another shared representation. The working assumption is that $\mathcal{Z}$ is sufficiently informative for an effective interval-level dynamics on $[0, 1]$.

4.3 Perturbation-indexed single-cell observations

For each perturbation $g \in \mathcal{G}$, let

$$X_g = \{x_{g1}, \dots, x_{gn_g^0}\} \subset \mathcal{Z}$$

be the latent states of observed *P4* cells carrying perturbation g , and let

$$Y_g = \{y_{g1}, \dots, y_{gn_g^1}\} \subset \mathcal{Z}$$

be the latent states of observed *P60* cells carrying perturbation g .

Define the empirical probability measures

$$p_0^g = \frac{1}{n_g^0} \sum_{i=1}^{n_g^0} \delta_{x_{gi}}, \quad p_1^g = \frac{1}{n_g^1} \sum_{j=1}^{n_g^1} \delta_{y_{gj}}. \quad (8)$$

4.4 Guide-abundance masses and finite endpoint measures

The numbers of sequenced cells and the guide-abundance measurements need not be on the same scale. Accordingly, we distinguish:

- *state samples*: n_g^0 and n_g^1 , used to define p_0^g and p_1^g ;
- *guide-abundance masses*: $M_g^0 > 0$ and $M_g^1 > 0$, representing relative perturbation abundance at *P4* and *P60* on the guide-abundance scale.

The initial and observed terminal finite measures are

$$\mu_0^g = M_g^0 p_0^g, \quad \nu_1^g = M_g^1 p_1^g. \quad (9)$$

This is the key endpoint object. For each perturbation g , the observed terminal target is the *finite measure* ν_1^g , which already combines:

- terminal geometry in latent state space through p_1^g , and
- terminal abundance on the guide-abundance scale through M_g^1 .

4.5 Optional interpretable tissue summaries

If one wants a biologically interpretable mean-field context, one may define:

$$\omega : \mathcal{Z} \rightarrow \Delta^{K-1}, \quad \phi : \mathcal{Z} \rightarrow \mathbb{R}^J,$$

where ω is an optional coarse population membership map and ϕ is an optional mediator or signaling feature map.

For each perturbation g , the observed terminal coarse composition is then

$$q_{g,1}^{\text{obs}} = \frac{1}{n_g^1} \sum_{j=1}^{n_g^1} \omega(y_{gj}) = \frac{1}{M_g^1} \int_{\mathcal{Z}} \omega(z) \nu_1^g(\mathrm{d}z). \quad (10)$$

These summaries are useful for interpretation, but they are not required for the core endpoint loss. The core supervision acts directly on finite measures.

5 Assumptions

The formulation rests on the following explicit assumptions.

- (A1) **Latent Markov sufficiency.** The pooled latent representation $\mathcal{Z}$ is sufficiently informative that the $P4 \rightarrow P60$ interval can be described by an effective Markovian dynamics on $\mathcal{Z}$.
- (A2) **Within-perturbation exchangeability.** For each perturbation g , conditional on latent state and the shared tissue context, the joint law of cells carrying g is invariant under permutation of cell indices. Thus, cells with the same perturbation are modeled as interchangeable particles, and the perturbation-specific empirical measure is the sufficient population-level state.
- (A3) **Relative-mass interpretation.** The masses M_g^0 and M_g^1 are treated as guide-abundance-scale masses and are therefore comparable across perturbations after normalization, even though they are distinct from the number of sequenced single-cell transcriptomes.
- (A4) **Mean-field closure.** The effect of the global tissue context on any cell is approximated through low-dimensional aggregate summaries of the joint population rather than through explicit pairwise interactions between all individual cells.
- (A5) **Control anchoring.** Control perturbations define the baseline continuation law. Perturbation effects are represented as low-dimensional deviations away from that baseline rather than as fully unrelated dynamics for each perturbation.
- (A6) **Effective-interval identifiability.** Because only two state-resolved time points are used, the learned dynamics are interpreted as an effective generator on $[0, 1]$, not as a uniquely identified microscopic law at all intermediate times.

6 Control anchoring and mean-field tissue context

6.1 Control perturbations and perturbation embeddings

Let

$$\mathcal{G}_{\text{ctrl}} \subset \mathcal{G}$$

denote the set of control perturbations. Each perturbation $g \in \mathcal{G}$ is assigned an embedding

$$a_g \in \mathbb{R}^r.$$

To make the control anchor exact, we impose

$$a_g = 0 \quad \text{for every } g \in \mathcal{G}_{\text{ctrl}}. \quad (11)$$

If several sgRNAs target the same gene, one may optionally introduce

$$a_{gr} = a_g + \delta_{gr}, \quad \delta_{gr} \in \mathbb{R}^r, \quad (12)$$

with shrinkage of δ_{gr} toward zero. For the core formulation below, a single perturbation-level embedding a_g is sufficient.

6.2 Joint population and total mass

Let

$$\mu_t = (\mu_t^g)_{g \in \mathcal{G}}$$

denote the family of perturbation-indexed populations at time t . The total population measure is

$$\bar{\mu}_t = \sum_{g \in \mathcal{G}} \mu_t^g, \quad \bar{M}(t) = \bar{\mu}_t(\mathcal{Z}). \quad (13)$$

6.3 Generic mean-field context

The most general formulation uses a low-dimensional tissue-context vector

$$c_t = \Psi_{\theta_c}(\mu_t) \in \mathbb{R}^L, \quad (14)$$

where Ψ_{θ_c} is a context map from the current joint population law to a low-dimensional summary.

This expression is the precise mathematical content of the mean-field coupling: the coefficient fields are allowed to depend on the current population only through the context vector c_t .

6.4 One interpretable choice of context map

One interpretable implementation is to define

$$q_t = \frac{1}{\bar{M}(t)} \sum_{g \in \mathcal{G}} \int_{\mathcal{Z}} \omega(z) \mu_t^g(\mathrm{d}z) \in \Delta^{K-1}, \quad (15)$$

and

$$s_t = \frac{1}{\bar{M}(t)} \sum_{g \in \mathcal{G}} \int_{\mathcal{Z}} \phi(z) \mu_t^g(\mathrm{d}z) \in \mathbb{R}^J. \quad (16)$$

Then set

$$c_t = \Psi_{\theta_c}(q_t, s_t) \in \mathbb{R}^L. \quad (17)$$

This implementation keeps the mean-field layer interpretable in terms of coarse population composition and mediator features, but the core formulation does not require this particular choice.

6.5 Time embedding and common network input

Let

$$\gamma : [0, 1] \rightarrow \mathbb{R}^{d_t}$$

be a time embedding. For a latent state $z \in \mathcal{Z}$, define

$$u_t(z; \mu_t) = (z, \gamma(t), c_t). \quad (18)$$

This is the common input to the coefficient networks.

7 Explicit control-anchored three-network weighted SDE

7.1 Why three separate networks

The conceptual model in [equations \(5\) and \(6\)](#) has three biologically distinct mechanisms:

- (i) directional movement,
- (ii) stochastic dispersion,
- (iii) multiplicative growth or depletion.

These mechanisms have different mathematical constraints and different identifiability properties. For that reason, they should not be collapsed into a single undifferentiated neural output. We therefore use:

- (1) a *drift network*,
- (2) a *dispersion network*,
- (3) a *growth network*.

7.2 Drift network: directional movement

Let

$$\text{DriftNet}_{\theta_b} : \mathbb{R}^{d+d_t+L} \rightarrow \mathbb{R}^d \times \mathbb{R}^{d \times r},$$

and write

$$\text{DriftNet}_{\theta_b}(u) = (\beta_b(u), B_b(u)),$$

where

$$\beta_b(u) \in \mathbb{R}^d, \quad B_b(u) \in \mathbb{R}^{d \times r}.$$

The perturbation-conditioned drift field is

$$b_g(z, t; \mu_t) = \beta_b(u_t(z; \mu_t)) + B_b(u_t(z; \mu_t)) a_g. \quad (19)$$

The first term is the baseline control law; the second term is the perturbation-specific deviation from control.

7.3 Dispersion network: stochastic spread

Let

$$\text{DispersionNet}_{\theta_\sigma} : \mathbb{R}^{d+d_t+L} \rightarrow \mathbb{R}^d \times \mathbb{R}^{d \times r},$$

and write

$$\text{DispersionNet}_{\theta_\sigma}(u) = (\beta_\sigma(u), B_\sigma(u)),$$

where

$$\beta_\sigma(u) \in \mathbb{R}^d, \quad B_\sigma(u) \in \mathbb{R}^{d \times r}.$$

Define the pre-activation

$$\ell_g(z, t; \mu_t) = \beta_\sigma(u_t(z; \mu_t)) + B_\sigma(u_t(z; \mu_t)) a_g \in \mathbb{R}^d, \quad (20)$$

and then define the diffusion matrix by

$$\sigma_g(z, t; \mu_t) = \text{diag}(\text{softplus}(\ell_g(z, t; \mu_t)) + \sigma_{\min}), \quad \sigma_{\min} > 0. \quad (21)$$

The diagonal form is the default because diffusion is only weakly identifiable from two endpoint snapshots. If additional information becomes available, one may replace this with a low-rank or full-covariance parameterization.

7.4 Growth network: expansion or depletion

Let

$$\text{GrowthNet}_{\theta_r} : \mathbb{R}^{d+d_t+L} \rightarrow \mathbb{R} \times \mathbb{R}^{1 \times r},$$

and write

$$\text{GrowthNet}_{\theta_r}(u) = (\beta_r(u), B_r(u)),$$

where

$$\beta_r(u) \in \mathbb{R}, \quad B_r(u) \in \mathbb{R}^{1 \times r}.$$

Define the unconstrained log-rate

$$\tilde{r}_g(z, t; \mu_t) = \beta_r(u_t(z; \mu_t)) + B_r(u_t(z; \mu_t)) a_g, \quad (22)$$

and then set

$$r_g(z, t; \mu_t) = r_{\max} \tanh(\tilde{r}_g(z, t; \mu_t)), \quad r_{\max} > 0. \quad (23)$$

The bounded tanh form stabilizes training because the particle weights depend exponentially on the time integral of r_g .

7.5 Exact control anchor

Because controls satisfy $a_g = 0$, [equations \(19\), \(20\) and \(22\)](#) imply that control perturbations follow the baseline continuation law produced by

$$\beta_b(\cdot), \quad \beta_\sigma(\cdot), \quad \beta_r(\cdot).$$

7.6 Particle system

For each perturbation g and each observed $P4$ cell x_{gi} , define a stochastic trajectory $Z_t^{g,i}$ and a positive weight $w_t^{g,i}$ by

$$dZ_t^{g,i} = b_g(Z_t^{g,i}, t; \mu_t) dt + \sigma_g(Z_t^{g,i}, t; \mu_t) dW_t^{g,i}, \quad Z_0^{g,i} = x_{gi}, \quad (24)$$

$$\frac{d}{dt} \log w_t^{g,i} = r_g(Z_t^{g,i}, t; \mu_t), \quad w_0^{g,i} = 1. \quad (25)$$

Here $W_t^{g,i}$ is a d -dimensional Brownian motion.

Therefore

$$w_t^{g,i} = \exp\left(\int_0^t r_g(Z_s^{g,i}, s; \mu_s) ds\right). \quad (26)$$

7.7 Population measure induced by the particle system

The perturbation-specific population measure is

$$\mu_t^g = \frac{M_g^0}{n_g^0} \sum_{i=1}^{n_g^0} w_t^{g,i} \delta_{Z_t^{g,i}} \in \mathcal{M}_+(\mathcal{Z}). \quad (27)$$

In particular, the predicted terminal mass is

$$\hat{M}_g^1 = \mu_1^g(\mathcal{Z}) = \frac{M_g^0}{n_g^0} \sum_{i=1}^{n_g^0} w_1^{g,i}. \quad (28)$$

This weighted empirical measure is the central predicted object.

7.8 Weak measure form

For every smooth test function $f : \mathcal{Z} \rightarrow \mathbb{R}$,

$$\frac{d}{dt} \int_{\mathcal{Z}} f(z) \mu_t^g(dz) = \int_{\mathcal{Z}} \left[b_g(z, t; \mu_t) \cdot \nabla f(z) + \frac{1}{2} \text{tr}(D_g(z, t; \mu_t) \nabla^2 f(z)) + r_g(z, t; \mu_t) f(z) \right] \mu_t^g(dz), \quad (29)$$

where

$$D_g(z, t; \mu_t) = \sigma_g(z, t; \mu_t) \sigma_g(z, t; \mu_t)^\top.$$

If a density exists, [equation \(29\)](#) corresponds formally to

$$\partial_t \mu_t^g = -\nabla \cdot (b_g(\cdot, t; \mu_t) \mu_t^g) + \frac{1}{2} \nabla \cdot \nabla : (D_g(\cdot, t; \mu_t) \mu_t^g) + r_g(\cdot, t; \mu_t) \mu_t^g. \quad (30)$$

7.9 Regularity conditions

For well-posedness of the particle system, we assume:

- (R1) the outputs of the three networks are measurable in (t, μ) and locally Lipschitz in z ;
- (R2) $\sigma_{\min} > 0$ and $r_{\max} < \infty$ are fixed constants;
- (R3) whenever interpretable summaries ω and ϕ are used, they are bounded.

8 Explicit details behind the loss function

8.1 Why the loss must act on finite measures

The forward model predicts, for each perturbation g , one terminal finite measure

$$\mu_1^g.$$

The observed endpoint is also one finite measure

$$\nu_1^g = M_g^1 p_1^g.$$

Therefore the statistically coherent supervision compares

$$\mu_1^g \quad \text{to} \quad \nu_1^g$$

directly as finite measures.

If one instead splits supervision into:

- a normalized state-distribution loss on $\hat{p}_1^g$, and
- a separate scalar loss on $\hat{M}_g^1$,

one artificially decomposes one endpoint object into two losses. In the present formulation, terminal geometry and terminal abundance are fitted jointly by a single unbalanced discrepancy.

8.2 Derived endpoint summaries

Although the primary predicted object is μ_1^g , it is often useful to report

$$\hat{M}_g^1 = \mu_1^g(\mathcal{Z}), \quad \hat{p}_1^g = \frac{\mu_1^g}{\hat{M}_g^1}, \quad (31)$$

and, when ω is used,

$$\hat{q}_{g,1} = \frac{1}{\hat{M}_g^1} \int_{\mathcal{Z}} \omega(z) \mu_1^g(dz) \in \Delta^{K-1}. \quad (32)$$

These are summaries of the finite measure, not replacements for it.

8.3 Generalized KL divergence for finite measures

For finite nonnegative measures $\alpha, \beta \in \mathcal{M}_+(\mathcal{Z})$, define the generalized KL divergence by

$$\text{KL}(\alpha \parallel \beta) = \begin{cases} \int_{\mathcal{Z}} \log\left(\frac{d\alpha}{d\beta}\right) \alpha(dz) - \alpha(\mathcal{Z}) + \beta(\mathcal{Z}), & \alpha \ll \beta, \\ +\infty, & \text{otherwise.} \end{cases} \quad (33)$$

This is the correct KL notion when the compared objects need not have the same total mass.

8.4 Entropic unbalanced optimal transport

Let $c(z, z')$ be the latent ground cost, typically

$$c(z, z') = \|z - z'\|_2^2.$$

For finite measures $\alpha, \beta \in \mathcal{M}_+(\mathcal{Z})$, define

$$\text{UOT}_{\varepsilon, \tau}(\alpha, \beta) = \min_{\pi \in \mathcal{M}_+(\mathcal{Z} \times \mathcal{Z})} \left\{ \int_{\mathcal{Z} \times \mathcal{Z}} c(z, z') \pi(\mathrm{d}z, \mathrm{d}z') + \varepsilon \text{KL}(\pi \parallel \alpha \otimes \beta) + \tau \text{KL}(\pi_1 \parallel \alpha) + \tau \text{KL}(\pi_2 \parallel \beta) \right\}, \quad (34)$$

where π_1, π_2 are the marginals of π .

This objective has four parts:

- (i) the transport term penalizes moving mass too far in latent space;
- (ii) the entropic term stabilizes the problem numerically;
- (iii) the marginal KL terms allow creation or destruction of mass relative to strict balanced transport;
- (iv) the parameter τ controls how costly that mass variation is.

8.5 Debiased unbalanced Sinkhorn divergence

To reduce entropic bias, define

$$\text{SD}_{\varepsilon, \tau}^{\text{unb}}(\alpha, \beta) = \text{UOT}_{\varepsilon, \tau}(\alpha, \beta) - \frac{1}{2} \text{UOT}_{\varepsilon, \tau}(\alpha, \alpha) - \frac{1}{2} \text{UOT}_{\varepsilon, \tau}(\beta, \beta). \quad (35)$$

8.6 Primary endpoint loss

The primary endpoint loss is

$$\mathcal{L}_{\text{end}} = \sum_{g \in \mathcal{G}} \alpha_g^{\text{end}} \text{SD}_{\varepsilon, \tau}^{\text{unb}}(\mu_1^g, \nu_1^g), \quad (36)$$

where α_g^{end} are user-chosen weights.

This loss is the central reason to formulate the endpoint as a finite-measure problem. It jointly penalizes:

- mismatch in terminal state geometry,
- mismatch in total terminal mass,
- and mismatch in the tradeoff between latent movement and mass change.

8.7 Optional auxiliary composition diagnostic

If one wants a weak interpretable diagnostic, one may define

$$\mathcal{L}_{\text{comp}}^{\text{aux}} = \sum_{g \in \mathcal{G}} \alpha_g^{\text{comp}} \text{KL}(q_{g,1}^{\text{obs}} \parallel \hat{q}_{g,1}). \quad (37)$$

Because $q_{g,1}^{\text{obs}}$ is derived from the same endpoint single-cell data that define p_1^g , this term is optional and not part of the primary supervision.

9 Regularization and identifiability

9.1 Embedding shrinkage

For non-control perturbations,

$$\mathcal{R}_{\text{emb}} = \sum_{g \in \mathcal{G} \setminus \mathcal{G}_{\text{ctrl}}} \|a_g\|^2. \quad (38)$$

If [equation \(12\)](#) is used, add

$$\sum_g \sum_{r \in \mathcal{R}_g} \|\delta_{gr}\|^2. \quad (39)$$

9.2 Perturbation-modulation regularization

Let

$$\tilde{p}_t^g = \frac{\mu_t^g}{\mu_t^g(\mathcal{Z})}$$

denote the normalized predicted population of perturbation g at time t . A natural penalty on perturbation-specific deviations from control is

$$\mathcal{R}_{\text{mod}} = \sum_{g \in \mathcal{G} \setminus \mathcal{G}_{\text{ctrl}}} \int_0^1 \mathbb{E}_{z \sim \tilde{p}_t^g} \left[\|B_b(u_t(z; \mu_t))a_g\|^2 + \|B_\sigma(u_t(z; \mu_t))a_g\|^2 + |B_r(u_t(z; \mu_t))a_g|^2 \right] dt. \quad (40)$$

9.3 Dispersion regularization

Because diffusion is only weakly identifiable from two endpoint snapshots, the model benefits from discouraging excessive stochastic spread:

$$\mathcal{R}_{\text{disp}} = \sum_{g \in \mathcal{G}} \int_0^1 \mathbb{E}_{z \sim \tilde{p}_t^g} \left[\|\sigma_g(z, t; \mu_t)\|_F^2 \right] dt. \quad (41)$$

9.4 Network parameter regularization

A generic parameter penalty is

$$\mathcal{R}_{\text{nn}} = \|\theta_b\|^2 + \|\theta_\sigma\|^2 + \|\theta_r\|^2 + \|\theta_c\|^2. \quad (42)$$

9.5 Why strong regularization is necessary

From only two state-resolved time points, drift, diffusion, and growth are not equally identifiable. In practice:

- drift carries the main burden of explaining large-scale state change;
- diffusion should remain relatively simple and strongly regularized;
- growth must be bounded because it exponentiates into the particle weights.

These are not merely numerical choices; they are part of the effective interval modeling assumptions.

10 Full training objective

The primary objective is

$$\mathcal{L} = \lambda_{\text{end}} \mathcal{L}_{\text{end}} + \lambda_{\text{emb}} \mathcal{R}_{\text{emb}} + \lambda_{\text{mod}} \mathcal{R}_{\text{mod}} + \lambda_{\text{disp}} \mathcal{R}_{\text{disp}} + \lambda_{\text{nn}} \mathcal{R}_{\text{nn}} + \lambda_{\text{comp}}^{\text{aux}} \mathcal{L}_{\text{comp}}^{\text{aux}}. \quad (43)$$

The default main formulation takes

$$\lambda_{\text{comp}}^{\text{aux}} = 0$$

or very small.

11 Numerical approximation and training

11.1 Euler–Maruyama approximation

Let

$$t_\ell = \ell \Delta t, \quad \Delta t = \frac{1}{L}, \quad \ell = 0, \dots, L-1.$$

Using [equations \(24\)](#) and [\(25\)](#), a standard Euler–Maruyama update is

$$Z_{\ell+1}^{g,i} = Z_\ell^{g,i} + b_g(Z_\ell^{g,i}, t_\ell; \mu_{t_\ell}) \Delta t + \sigma_g(Z_\ell^{g,i}, t_\ell; \mu_{t_\ell}) \sqrt{\Delta t} \xi_\ell^{g,i}, \quad (44)$$

$$w_{\ell+1}^{g,i} = w_\ell^{g,i} \exp\left(r_g(Z_\ell^{g,i}, t_\ell; \mu_{t_\ell}) \Delta t\right), \quad (45)$$

where

$$\xi_\ell^{g,i} \sim \mathcal{N}(0, I_d).$$

11.2 Context update

If the interpretable context construction [equations \(15\)](#) to [\(17\)](#) is used, then at each step one recomputes

$$q_{t_\ell} = \frac{1}{\bar{M}(t_\ell)} \sum_{g \in \mathcal{G}} \frac{M_g^0}{n_g^0} \sum_{i=1}^{n_g^0} w_\ell^{g,i} \omega(Z_\ell^{g,i}), \quad (46)$$

$$s_{t_\ell} = \frac{1}{\bar{M}(t_\ell)} \sum_{g \in \mathcal{G}} \frac{M_g^0}{n_g^0} \sum_{i=1}^{n_g^0} w_\ell^{g,i} \phi(Z_\ell^{g,i}), \quad (47)$$

and then set

$$c_{t_\ell} = \Psi_{\theta_c}(q_{t_\ell}, s_{t_\ell}). \quad (48)$$

If a different context map is used, one updates c_{t_ℓ} according to that construction instead.

11.3 Discrete terminal measures

At the terminal step $t_L = 1$, the predicted finite measure is

$$\mu_1^g = \frac{M_g^0}{n_g^0} \sum_{i=1}^{n_g^0} w_L^{g,i} \delta_{Z_L^{g,i}}, \quad (49)$$

while the observed terminal measure is

$$\nu_1^g = \frac{M_g^1}{n_g^1} \sum_{j=1}^{n_g^1} \delta_{y_{gj}}. \quad (50)$$

The discrete unbalanced OT problem underlying [equation \(36\)](#) is then built from the cost matrix

$$C_{ij}^g = c(Z_L^{g,i}, y_{gj}).$$

11.4 Practical training loop

A practical training loop is:

- (1) construct the shared latent space $\mathcal{Z}$ from pooled $P4$ and $P60$ data;
- (2) initialize particles at the observed $P4$ cells for each perturbation;
- (3) jointly simulate all perturbation-indexed populations from $t = 0$ to $t = 1$ using [equations \(44\)](#) and [\(45\)](#);
- (4) compute the terminal measures μ_1^g and the derived summaries $\hat{M}_g^1$, $\hat{p}_1^g$, and optionally $\hat{q}_{g,1}$;
- (5) evaluate the primary endpoint loss [equation \(36\)](#) and the regularizers [equations \(38\)](#) and [\(40\)](#) to [\(42\)](#);
- (6) optionally evaluate [equation \(37\)](#) for diagnosis or weak auxiliary regularization;
- (7) update parameters by stochastic gradient descent.

If jointly simulating all perturbations is too costly, one may use a stratified minibatch of perturbations to estimate the mean-field terms. That changes the numerical estimator, not the population-level model.

12 Biological outputs and counterfactuals

The formulation produces the following outputs.

12.1 Predicted terminal finite measure

The central endpoint object is

$$\mu_1^g,$$

which jointly encodes terminal abundance and terminal state allocation.

12.2 Predicted abundance effect

The abundance or selection effect of perturbation g is summarized by

$$\hat{M}_g^1 \quad \text{or} \quad \log \hat{M}_g^1 - \log M_g^0.$$

12.3 Predicted terminal state distribution

The normalized endpoint state distribution is

$$\hat{p}_1^g.$$

12.4 Predicted terminal coarse composition

If ω is used, the predicted coarse endpoint allocation is

$$\hat{q}_{g,1}.$$

This is an interpretive summary, not the main fitted target.

12.5 Intrinsic versus context-mediated effect

Because the model separates perturbation embedding from tissue context, one may compare:

- (a) a full simulation using the self-consistent mean field c_t , and
- (b) a simulation with c_t clamped to a reference trajectory or removed.

This gives a principled decomposition of endpoint effects into intrinsic perturbation components and context-mediated components.

12.6 Control-referenced perturbation effect

Because controls correspond to $a_g = 0$, one may also compare:

- (a) evolution under a_g , and
- (b) evolution under $a = 0$,

starting from the same initial state distribution. This directly quantifies the perturbation effect relative to the control continuation law.

13 What this model is, and is not

13.1 It is a mean-field interaction model

Mathematically, the model is a perturbation-indexed McKean–Vlasov reaction–drift–diffusion system with low-dimensional population feedback through the tissue-context vector c_t .

13.2 It is not a full mean-field game

A literal mean-field game would require explicit control variables, value functions, and a coupled Hamilton–Jacobi–Bellman / Fokker–Planck system. None of those are introduced here. The model uses mean-field feedback, but it is not a full equilibrium game.

13.3 It estimates an effective interval generator

From two state-resolved time points, the continuous-time law between them is not uniquely identifiable. The learned coefficient fields must therefore be interpreted as a regularized effective generator on $[0, 1]$, not as the unique microscopic biological truth.

13.4 It supports intervention-conditional interpretation, not unrestricted causal claims

The model is suitable for counterfactual simulation within the support of the observed perturbation-conditioned data and latent representation. It does not, by itself, identify unrestricted causal effects outside that support.

14 Concise summary

The formulation can be summarized compactly as follows:

For each perturbation g , start from the observed $P4$ finite measure $\mu_0^g = M_g^0 p_0^g$, evolve a weighted particle system in latent space under a control-anchored perturbation-conditioned McKean–Vlasov neural SDE whose drift, stochastic dispersion, and growth are modeled by three separate neural networks conditioned on latent state, time, and low-dimensional tissue context, and fit the resulting terminal finite measure μ_1^g to the observed $P60$ finite measure $\nu_1^g = M_g^1 p_1^g$ by a single unbalanced optimal-transport discrepancy.

Equivalently, the model has five layers:

- (1) a *biological mechanism layer*, which separates movement, stochastic spread, and growth/depletion;
- (2) an *SDE layer*, which evolves latent cell states;
- (3) a *weight layer*, which evolves perturbation-specific mass;
- (4) a *mean-field context layer*, which feeds back low-dimensional tissue information;
- (5) an *endpoint loss layer*, which compares predicted and observed terminal finite measures by unbalanced optimal transport.

This is the explicit formal version of the $P4 \rightarrow P60$ control-anchored mean-field neural differential equation formulation after placing the biology first, explicitly introducing the SDE, and then giving the detailed finite-measure model and loss function.